

EXAM 7 – SPRING 2013

8. (5.75 points)

Given the following information for an insurance company's outstanding claim liabilities as of December 31, 2012:

<u>Coefficients of Variation (CoV)</u>				
<u>Line of business</u>	<u>Central Estimate</u>	<u>Independent risk</u>	<u>Internal systemic risk</u>	<u>External systemic risk</u>
Workers compensation	\$110,000,000	7.2%	7.5%	3.4%
Auto	95,000,000	5.2%	4.5%	4.3%
Homeowners	<u>45,000,000</u>	5.8%	5.0%	<u>6.7%</u>
Total	250,000,000			4.7%

- Assume that the claim liabilities are normally distributed.
- The z-value for the 75th percentile of the normal distribution is: 0.674
- The selected correlation between each pair of lines of business for internal systemic risk is: 0.25
- Assume that each source of independent risk is uncorrelated with any other source of risk.

a. (0.5 point)

Calculate the independent risk CoV for all lines combined.

b. (0.5 point)

Describe why it is reasonable to assume that there is positive correlation between lines of business for internal systemic risk.

c. (1 point)

Calculate the internal systemic risk CoV for all lines combined.

d. (0.5 point)

Calculate the consolidated CoV for all sources of risk.

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(8 continued)

e. (0.5 point)

Calculate the risk margin (in dollars) for the overall book of business that provides a 75% probability of reserve adequacy.

f. (0.5 point)

Discuss how the risk margin in e. above would change if the selected pair-wise correlation between each pair of lines of business for internal systemic risk were higher than 0.25.

g. (1.5 points)

Analyze the consistency of the selected CoVs by line of business using internal benchmarking checks. Discuss all three main sources of risk.

h. (0.75 point)

Discuss an external systemic risk for which the assumption of constant correlation between lines of business across their loss distributions does not hold. Suggest a method to incorporate variable correlation into the risk margin.

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Question 8 Sample Solution

Solution 1

- a) The Φ of independent risk = $(\sum (\omega_i \times \Phi_i)^2)^{0.5}$;
= $[(.44 \times 0.72)^2 + (.38 \times .052)^2 + (.18 \times .058)^2]^{0.5}$;
= .039
- b) In the case of the same individual performing actuarial analyses of the central estimates for all lines of business, the “same actuary effect” can be present, resulting in estimates that are subject to the same biases and therefore positively correlated.
- c) $\Phi_{\text{Internal}} = [\sum \rho_{ij} \times (\omega_i \times \Phi_i)(\omega_j \times \Phi_j)]^{0.5}$ where $\rho_{ij} = 0.25$ for all i, j
= $[.44 \times 0.75 \times (1 \times .44 \times .075 + .25 \times .38 \times .045 + .25 \times .18 \times .05) \text{ --- WC}$
+ $.38 \times .045 \times (.25 \times .44 \times .075 + 1 \times .38 \times .045 + .25 \times .18 \times .05) \text{ --- Auto}$
+ $.18 \times .05 \times (.25 \times .44 \times .075 + .25 \times .38 \times .045 + 1 \times .18 \times .05)]^{0.5} \text{ --- HO}$
= .044
- d) $\Phi_{\text{External}} = .047$ (given in problem)
 $\Phi_{\text{Total}} = [.039^2 + .044^2 + .047^2]^{0.5}$
= .075
- e) Risk margin = $Z_\alpha \times \Phi_{\text{Total}} = .674 \times .075 = .05055 \times \$250\text{m} = \$12,637,500$
- f) A high cross correlation would raise the internal risk CoV, since adverse results of one class would mean adverse results in another class with a stronger dependency. The overall CoV would then increase as well as the risk margin.
- g) 1-Independent Risk: This includes both process risk (randomness) and parameter risk, the risk that parameters selected do not fit the line of business. The CoV's rank from least complex line (auto) to most complex (work comp), and it makes sense that a more complex line such as work comp, which involves additional parameters (inflation, court decisions, disability schedules) would have a higher CoV for independent risk.
- 2-Internal Systemic Risk: This includes specification error, parameter selection error, and data error. Among the risks here are the same actuary effect, lack of knowledge about a line of business, and the idea that no model will accurately capture the entire risk. The resulting CoVs follow a similar pattern as independent risk, that is a less complex line (auto) has a lower CoV than work comp. The follows the logic discussed above.

3-External Systemic Risk: This includes economic and social risk, legislative, political and claims inflation risk, claims management process risk, event risk, expense risk, latent claim risk, and recovery risk. HO has the highest CoV here, likely due to the influence of event risk on this book of business. Also, since HO is a smaller liability than work comp or auto, its CoV will be higher here. It is surprising that auto's CoV is higher than work comp, though this may also be due to event risk and due to a smaller liability size than WC.

- h) Event risk – There will be increased correlation in the tail between auto and homeowners in the event of a catastrophe (and work comp as well if it is an earthquake with damage to a building while employees are in it). To incorporate this variable correlation, a copula could be used to join the distributions of different lines of business. This would be more rigorous computationally, but would result in a more accurate view of increased tail dependencies in the case of a catastrophic event.

Solution 2

a) $\text{Total cov}^2 = \sum (\omega_i \times \Phi_i)^2$

$$= \{ [(110/250) \times .075]^2 + [(95/250) \times .052]^2 + [(45/250) \times .058]^2 \}^{0.5}$$

$$= .0388$$

- b) Internal systemic risk is risk internal to the insurance liability valuation modeling process. Possible risks include data errors, parameter mis-specification, or even the “same actuary” effect, all of which will impact multiple lines of business in the same way thus resulting in positive correlation.

c) $[\sum \sum \rho_{ij} \times (\omega_i \times \Phi_i)(\omega_j \times \Phi_j)]^{0.5}$

$$= [(110/250 \times .075)^2 + (95/250 \times .045)^2 + (45/250 \times .05)^2$$

$$+ 2 \times (.25) \times (110/250 \times .075)(95/250 \times .045) \quad \leftarrow \text{WC + Auto}$$

$$+ 2 \times (.25) \times (110/250 \times .075)(45/250 \times .05) \quad \leftarrow \text{WC + HO}$$

$$+ 2 \times (.25) \times (95/250 \times .045)(45/250 \times .05)]^{0.5} \quad \leftarrow \text{Auto + HO}$$

$$= [.00146 + .00028 + .000149 + .000077]^{0.5}$$

$$= .0444$$

d) $= [.0388^2 + .0444^2 + .047^2]^{0.5} = .0754$

e) $= \mu \times (1 + Z \times \text{CoV}) - \mu = \text{Risk Margin}$

$$250 \times (1 + .674 \times .0754) - 250 = 12,705,615$$

- f) If the correlations were higher, this would increase the overall CoV for internal systemic risk, which would increase the total CoV and thus increase the risk margin.
- g) For independent risk, we should consider the portfolio size and length of claim runoff when selecting CoV's. As we might expect, WC has a higher CoV because it has a longer claim runoff. This should be tempered though by the fact that it is the largest in size of reserves – law of large numbers should drive this CoV down.

For internal systemic risk, the low CoV's for HO and auto suggest we are more confident in our models and parameters compared to WC. Perhaps we have fewer models for WC or haven't identified the best predictors of claim cost outcomes.

For external systemic risk, we might expect the HO CoV to be lower since it is for O/S claim liability only (the prem liability CoV would be higher due to event risk). The WC CoV seems low given the risk of claim inflation and the potential for legislative changes for this LOB.

- h) Event risk for auto and homeowners. Event risk is the risk of a single event causing a large number of claims. Auto and HO results are likely to be uncorrelated except in the tails of the distribution, for instance when there is a large catastrophe. We could use copulas to model this correlation structure so that the results only show dependency in tails.

Examiner Comment

Part a) of the problem depended upon an understanding that the weighted CoV values by line should be squared and then added, prior to taking the square root for the overall independent risk. Common errors included not using the line of business proportional weights in the formula, not squaring the weighted CoV values prior to taking their sum, or squaring the individual line CoV values but not squaring the individual line weights.

Part b) of the problem could be answered in several ways for full credit, however simply mentioning “the same actuary effect” without any elaboration did not qualify for full credit. A number of candidates referred to internal system issues that would affect multiple lines of business (e.g., policy issuance problems) but had no mention of uncertainty arising from actuarial valuation models and therefore that type of response did not receive any credit.

Part c) of the problem depended upon the proper application of the formula using weighted CoV values by lines of business and incorporating the effect of correlations between lines. Candidate errors on this part included the omission of line of business weights in the formula, failure to

include a factor of 2.0 in the correlation component of the formula for each line of business pair, and failure to square the line of business weights in the stand-alone CoV components for each line of business, all leading to minor point reductions. Part d) of the problem required the use of the CoV values calculated in parts a) and c) of the problem along with the total external systemic risk CoV value stated in the problem. If the candidate had miscalculated either or both of the CoV values in parts a) and c) of the problem but used the errant figures correctly in the setup to part d), there was no additional penalty applied in this part of the problem. Also, due to the complexity of the formula in this part, candidates were more likely to make a computational error in evaluating the formula value, but as long as the formula set-up was clearly shown only a minor penalty was applied for that computational error.

Part e) of the problem required the candidate to use the consolidated CoV value determined in part d) along with the Z-value given in the problem and calculate a risk margin in dollars. If a candidate had miscalculated the consolidated CoV in part d) of the problem but used the errant figure correctly in the setup to part e), there was no additional penalty applied in this part. Some candidates also failed to calculate the risk margin in dollars as explicitly required by the problem or else provided the low/high range of reserve values instead of a reserve margin, both of which resulted in only partial credit. Part f) of the problem required some explanation of how the risk margin in part e) would change; therefore candidate solutions that stated solely that the margin would increase received only partial credit.

Part g) of the problem required some discussion of the relative CoV values within each risk category (independent, internal systemic, and external systemic) for full credit, along with some recognition that the external systemic risk CoV for the homeowners line appears abnormally high relative to the other two lines of business. Part h) required the candidate to explain why the assumption of constant correlation between lines of business across their loss distributions does not hold. Some candidates mentioned event risk and noted that correlations would be higher in the tail but did not explain why or give any example which resulted in a minor point reduction.